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Subject : DSA

Unit 2, S6, SLO2 - Singly Linked List – Deletion at Beginning and End

Algorithm to delete the first node

Step 1: IF START = NULL

Write _____

Go to Step 5

[END OF IF]

Step 2: SET PTR = _____

Step 3: SET START = _____

Step 4: FREE _____

Step 5: EXIT

Answer: Step 1: Write UNDERFLOW

Step 2: SET PTR = START

Step 3: SET START = START -> NEXT

Step 4: FREE PTR

Step 5: EXIT

Algorithm to delete the last node

Step 1: IF _____

Write UNDERFLOW

Go to Step 8

[END OF IF]

Step 2: SET PTR = _____

Step 3: Repeat Steps 4 and 5 while PTR -> NEXT != _____

Step 4: SET _____ **= PTR**

Step 5: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 6: SET _____ **= NULL**

Step 7: FREE _____

Step 8: EXIT

Answer: Step 1: IF START = NULL

Step 2: SET PTR = START

Step 3: while PTR -> NEXT != NULL

Step 4: SET PREPTR = PTR

Step 5: SET PTR = PTR -> NEXT

Step 6: SET PREPTR -> NEXT = NULL

Step 7: FREE PTR

Step 8: EXIT